Pars Network

Sovereign Infrastructure for the Persian Diaspora

Version 1.0 — 2025

This is the Pars 1.0 flagship paper.
For the GPU-native sovereign L1
with DEX+EVM+FHE trifecta

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Abstract

The Iranian diaspora—estimated at eight to ten million individuals distributed across sixty countries—lacks sovereign digital infrastructure that is simultaneously censorship-resistant, culturally coherent, and economically self-sustaining. Existing platforms are subject to jurisdictional seizure, sanctions-regime over-compliance, and surveillance by both host-country and origin-country intelligence services. We present **Pars Network**, a dual-layer architecture combining an EVM-compatible Layer 2 rollup with a privacy-preserving Session daemon layer. The rollup layer provides programmable settlement with sub-dollar transaction costs and full Ethereum security inheritance. The Session layer delivers end-to-end encrypted communication, coercion-resistant governance, and zero-knowledge identity primitives purpose-built for communities operating under authoritarian threat models. We formalize the security properties required for diaspora sovereignty, introduce the Pars Session Protocol (PSP) for deniable encrypted communication, describe a post-quantum migration strategy, and present governance mechanisms resilient to both Sybil attacks and state-level coercion. Benchmarks on testnet demonstrate 2,400 TPS on the rollup layer with 1.2-second confirmation latency and sub-second message delivery on the Session layer with forward secrecy guarantees.

Keywords: diaspora infrastructure, censorship resistance, EVM rollup, encrypted communication, zero-knowledge identity, post-quantum cryptography, coercion resistance

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1 Introduction

The Persian-speaking diaspora constitutes one of the largest and most economically significant expatriate populations worldwide. Conservative estimates place the population at eight million, with concentrations in North America (3.5M), Europe (2.5M), the Persian Gulf states (1.2M), and Oceania (0.5M) [1]. Despite this scale, the community relies entirely on infrastructure controlled by entities whose incentives diverge from diaspora interests.

1.1 Problem Statement

Three structural failures define the current landscape:

(i) **Platform dependence.** Communication channels (Telegram, WhatsApp, Instagram) are operated by corporations subject to regulatory pressure from both Western sanctions regimes and Iranian surveillance apparatus. Telegram channels with millions of Persian-speaking subscribers have been seized or silenced without recourse [2].

(ii) **Financial exclusion.** Sanctions over-compliance by banks and payment processors systematically excludes Iranian nationals—including legal permanent residents and citizens of host countries—from financial services. The compliance cost of serving this population exceeds the revenue for most institutions [3].

(iii) **Identity fragmentation.** Diaspora members maintain multiple inconsistent identities across jurisdictions, with no portable credential system. Legal name changes, transliteration inconsistencies, and lack of apostille infrastructure create persistent friction [4].

1.2 Design Principles

Pars Network is architected around five invariant principles:

Definition 1 (Sovereignty Invariant)**.** *No single jurisdiction, corporation, or governance body may unilaterally halt, censor, or modify network operations affecting more than one percent of active participants.*

Definition 2 (Cultural Coherence)**.** *Protocol interfaces, governance processes, and user-facing systems must support right-to-left text rendering, Persian calendar systems, and Persian-language-first interaction without degraded functionality.*

Definition 3 (Coercion Resistance)**.** *Participation in network governance must be deniable. No on-chain or off-chain artifact may prove that an individual cast a specific vote or holds a specific identity credential.*

Definition 4 (Economic Self-Sufficiency)**.** *Network operations must be funded through protocol-level mechanisms (transaction fees, staking yields, storage rents) rather than external grants or corporate sponsorship.*

Definition 5 (Gradualist Security)**.** *The system must function under current cryptographic assumptions while maintaining a credible migration path to post-quantum security within a single protocol version upgrade.*

1.3 Contributions

This paper makes the following contributions:

- A dual-layer architecture combining EVM rollup settlement with an off-chain Session daemon for encrypted communication (Section 3).

- The Pars Session Protocol (PSP), providing deniable authentication with forward secrecy and post-quantum hybrid key exchange (Section 4).
- A coercion-resistant governance mechanism based on homomorphic tallying with receipt-freeness (Section 5).
- A zero-knowledge identity framework supporting selective disclosure of diaspora-relevant credentials (Section 6).
- A post-quantum migration strategy with hybrid classical/lattice-based cryptographic schemes (Section 7).
- Economic analysis and incentive design for network sustainability (Section 8).
- Testnet benchmarks demonstrating performance feasibility (Section 10).

2 Threat Model

We consider three adversary classes, each with distinct capabilities:

2.1 Nation-State Adversary (Origin Country)

Assumption 1 (NS-Origin)**.** *The origin-country adversary controls:*

- *All ISP-level traffic within its borders, with deep packet inspection capability.*
- *A network of human intelligence assets within diaspora communities.*
- *Legal mechanisms to compel family members remaining in-country.*
- *Computational resources estimated at* 10^{12} *AES-equivalent operations per second.*

2.2 Nation-State Adversary (Host Country)

Assumption 2 (NS-Host)**.** *Host-country adversaries control:*

- *Lawful intercept capabilities on domestic telecommunications.*
- *Subpoena power over domestically incorporated service providers.*
- *Sanctions enforcement mechanisms affecting financial transactions.*
- *Metadata collection programs operating at national scale.*

2.3 Sophisticated Non-State Adversary

Assumption 3 (Non-State)**.** *Non-state adversaries possess:*

- *Budget of \$10M–\$100M for targeted operations.*
- *Access to commercial spyware (e.g., Pegasus-class tools).*
- *Social engineering capabilities within community networks.*
- *Ability to operate compromised nodes within the network.*

2.4 Security Goals

Given these adversary classes, Pars Network must satisfy:

Table 1: Security goals and adversary mapping

Property	Description	NS-O	NS-H	NSA
Confidentiality	Message content hidden from non-recipients	✓	✓	✓
Metadata resistance	Communication patterns obscured	✓	Partial	✓
Censorship resistance	Messages delivered despite blocking	✓	N/A	✓
Coercion resistance	Votes/credentials deniable under duress	✓	✓	✓
Forward secrecy	Past messages secure if keys compromised	✓	✓	✓
Post-compromise security	Future messages secure after recovery	✓	✓	✓
Transaction privacy	Financial transfers unlinkable	Partial	✓	✓
Identity privacy	Real identity unlinkable to network identity	✓	✓	✓

3 Architecture Overview

Pars Network employs a dual-layer architecture that separates concerns between verifiable computation (Layer 1/L2) and private communication (Session Layer).

3.1 Layer 2 Rollup

The settlement layer is an optimistic rollup deployed on Ethereum, chosen for its security properties and existing Persian-speaking developer ecosystem.

Definition 6 (Pars Rollup)**.** *A tuple* $\mathcal{R} = (\mathcal{S}, \mathcal{T}, \delta, \pi, \mathcal{V})$ *where:*

- $\mathcal{S}$*: State space (account balances, contract storage, identity registry)*
- $\mathcal{T}$*: Transaction space (EVM transactions, identity operations, governance actions)*
- $\delta : \mathcal{S} \times \mathcal{T} \to \mathcal{S}$*: State transition function*
- π*: Proof system (fraud proofs for optimistic verification)*
- $\mathcal{V}$*: Verifier contract on Ethereum L1*

3.1.1 Sequencer Design

The rollup employs a decentralized sequencer set with the following properties:

Algorithm 1 Decentralized Sequencer Selection

Require: Validator set $V = \{v_1, \ldots, v_n\}$, stakes $\{s_1, \ldots, s_n\}$, epoch e
Ensure: Selected sequencer v^* for epoch e

```
seed ← VRF(epoch_seed(e))
total ← Σ_{i=1}^{n} s_i
r ← seed mod total
cumulative ← 0
for i = 1 to n do
    cumulative ← cumulative + s_i
    if r < cumulative then
        return v_i
    end if
end for
```

The sequencer rotation period is 256 blocks (approximately 5 minutes), ensuring no single sequencer controls ordering for extended periods.

3.1.2 Fraud Proof System

We employ interactive fraud proofs following the Arbitrum model [5], with modifications for identity-aware state transitions:

1. **Assertion.** Sequencer posts state root S_n with bond B.
2. **Challenge.** Any validator may challenge within period $\tau_c = 7$ days.
3. **Bisection.** Challenger and asserter bisect the disputed computation.
4. **Single-step proof.** The disputed step is executed on L1.
5. **Resolution.** Incorrect party's bond is slashed; correct party is rewarded.

3.1.3 EVM Compatibility

The execution environment is fully EVM-compatible with extensions for:

- `IDENTITY_VERIFY`: Precompile for on-chain ZK identity proof verification.
- `SESSION_ANCHOR`: Precompile for anchoring Session layer state commitments.
- `CALENDAR_CONVERT`: Precompile for Solar Hijri calendar operations.
- `RTL_HASH`: Precompile for deterministic right-to-left text normalization.

3.2 Session Daemon Layer

The Session layer operates as a peer-to-peer overlay network providing encrypted communication, identity management, and governance participation.

3.2.1 Node Architecture

Each Pars node runs three processes:

1. **Rollup client**: Syncs L2 state and submits transactions.
2. **Session daemon**: Manages encrypted sessions, message routing, and key material.
3. **Relay service**: Provides NAT traversal, store-and-forward, and Tor integration.

3.2.2 Cross-Layer Communication

The layers communicate through two mechanisms:

Definition 7 (Anchor Commitment)**.** *The Session daemon periodically commits a Merkle root of its state to the rollup via the* `SESSION_ANCHOR` *precompile:*

$$anchor(e) = MerkleRoot(sessions(e) \| keys(e) \| messages(e))$$

where e denotes the epoch number.

Definition 8 (Proof Bridge)**.** *Any Session-layer claim (message delivery, key rotation, identity attestation) can be verified on-chain by providing a Merkle inclusion proof against the most recent anchor commitment.*

3.3 Network Topology

The network organizes into three tiers:

Table 2: Network node tiers

Tier	Role	Stake Required	Expected Count
Validator	Sequence transactions, produce blocks	32,000 PARS	100–500
Relay	Route messages, provide NAT traversal	1,000 PARS	1,000–10,000
Light	Send/receive, verify proofs	0 PARS	100,000+

4 Pars Session Protocol

The Pars Session Protocol (PSP) provides end-to-end encrypted communication with properties specifically designed for the diaspora threat model.

4.1 Protocol Goals

1. **Confidentiality**: Messages readable only by intended recipients.
2. **Forward secrecy**: Compromise of long-term keys does not reveal past messages.
3. **Post-compromise security**: Sessions self-heal after temporary key compromise.
4. **Deniability**: No cryptographic proof that a specific user sent a specific message.
5. **Offline delivery**: Messages queued for offline recipients with bounded storage.
6. **Group support**: Efficient encrypted group communication for up to 10,000 members.

4.2 Key Hierarchy

PSP employs a four-level key hierarchy:

$$\text{Level 0:} \quad \text{Identity key pair } (ik_{pub}, ik_{priv}) \text{ — long-term, per-device} \tag{1}$$
$$\text{Level 1:} \quad \text{Signed pre-key } (spk_{pub}, spk_{priv}) \text{ — rotated monthly} \tag{2}$$
$$\text{Level 2:} \quad \text{Ephemeral pre-keys } \{(epk^i_{pub}, epk^i_{priv})\}_{i=1}^{N} \text{ — one-time use} \tag{3}$$
$$\text{Level 3:} \quad \text{Session keys } \{sk_j\}_{j=1}^{M} \text{ — ratcheted per message} \tag{4}$$

4.3 Session Establishment

Session establishment follows a modified X3DH protocol [6] with hybrid post-quantum key exchange:

Algorithm 2 Hybrid X3DH Session Establishment

Require: Alice's identity key ik_A, Bob's bundle (ik_B, spk_B, epk_B)
Ensure: Shared secret SS for session initialization

```
// Classical ECDH components
dh_1 ← ECDH(ik_A, spk_B)
dh_2 ← ECDH(ek_A, ik_B)
dh_3 ← ECDH(ek_A, spk_B)
dh_4 ← ECDH(ek_A, epk_B)
ss_classical ← KDF(dh_1 || dh_2 || dh_3 || dh_4)
// Post-quantum Kyber component
(ct, ss_pq) ← Kyber.Encaps(ik_B^kyber)
// Hybrid combination
SS ← KDF(ss_classical || ss_pq)
return (SS, ct)
```

4.4 Double Ratchet with PQ Enhancement

After session establishment, PSP employs a Double Ratchet [7] with post-quantum ratchet steps:

Algorithm 3 PQ-Enhanced Double Ratchet Step

Require: Current ratchet state $(rk, ck_s, ck_r, dh_{self}, dh_{remote})$
Ensure: Updated state, message key mk

```
if new DH ratchet public key received from remote then
    dh_remote ← received public key
    (rk', ck_r) ← KDF_RK(rk, ECDH(dh_self, dh_remote))
    Generate new DH key pair dh_self
    (rk, ck_s) ← KDF_RK(rk', ECDH(dh_self, dh_remote))
    if counter mod Q = 0 then                          ▷ Periodic PQ ratchet
        (ct, ss_pq) ← Kyber.Encaps(pk_remote^kyber)
        rk ← KDF(rk || ss_pq)
    end if
end if
(ck_s, mk) ← KDF_CK(ck_s)
return mk
```

The parameter Q controls the frequency of post-quantum ratchet steps; we set $Q = 50$ messages as a balance between computational overhead and quantum security margin.

4.5 Deniable Authentication

PSP achieves deniability through a designated verifier signature scheme:

Theorem 1 (Deniability)**.** *For any message m exchanged between Alice and Bob, there exists a simulator $\mathcal{S}$ such that for any verifier V:*

$$\Pr[V(transcript_{real}) = 1] = \Pr[V(\mathcal{S}(transcript_{sim})) = 1]$$

where $transcript_{real}$ is the actual protocol transcript and $transcript_{sim}$ is the simulated transcript.

Proof sketch. The key exchange in Algorithm 2 uses only Diffie-Hellman operations, which are inherently deniable: Bob can compute $dh_2 = \text{ECDH}(ek_A, ik_B)$ himself using his own identity key, thus simulating the entire transcript. The Kyber component uses CCA-secure encapsulation, and we add a commitment scheme where both parties can claim to have generated the ciphertext. Full proof in Appendix A. □

5 Coercion-Resistant Governance

Governance for diaspora communities must resist coercion by state actors who may physically compel individuals to reveal their votes or delegate their voting power.

5.1 Threat Model for Governance

Definition 9 (Coercion)**.** *An adversary $\mathcal{A}$ coerces voter v if $\mathcal{A}$ can:*

1. *Demand that v vote in a specific way, and*
2. *Verify whether v complied with the demand.*

Definition 10 (Receipt-Freeness)**.** *A voting scheme is receipt-free if no voter can construct a proof of how they voted, even if they wish to do so.*

5.2 Voting Protocol

We employ a modified JCJ protocol [8] adapted for blockchain-based tallying:

Algorithm 4 Coercion-Resistant Vote Casting

Require: Voter credential σ, vote v, election public key pk_E
Ensure: Encrypted ballot β posted on-chain

```
// Generate fresh credential
σ' ← Rerandomize(σ)
// Encrypt vote with election key
c_v ← ElGamal.Enc(pk_E, v; r_1)
// Encrypt credential
c_σ ← ElGamal.Enc(pk_E, σ'; r_2)
// ZK proof of valid vote and credential
π ← ZKProof{(v, σ', r_1, r_2) : c_v = Enc(v) ∧ c_σ = Enc(σ') ∧ σ' ∈ Credentials}
β ← (c_v, c_σ, π)
Post β to rollup via anonymous relay
return β
```

5.3 Panic Credentials

A novel contribution is the *panic credential* mechanism:

Definition 11 (Panic Credential)**.** *Each voter possesses two credentials: a* real credential σ_r *and a* panic credential σ_p*. Votes cast with σ_p are silently discarded during tallying. The credentials are computationally indistinguishable to any party other than the voter.*

Under coercion, a voter uses σ_p to cast the demanded vote. The coercer observes a valid ballot on-chain but the vote does not affect the outcome. The voter subsequently casts their true preference using σ_r through an anonymous channel.

5.4 Tallying

Tallying is performed by a threshold decryption committee using Shamir secret sharing:

1. Committee members jointly decrypt all ballots.
2. Ballots with panic credentials are identified and removed (only the committee can distinguish them via threshold decryption of the credential ciphertext).
3. Duplicate credentials are resolved by keeping only the latest ballot.
4. Final tally is published with a ZK proof of correct computation.

5.5 Governance Structure

Table 3: Pars governance bodies

Body	Responsibility	Members	Term
Assembly	Protocol upgrades, parameter changes	21	1 year
Council	Treasury management, grants	7	6 months
Guardians	Emergency actions, circuit breakers	5	2 years
Community	Proposal submission, referendum votes	All stakers	Continuous

5.6 Proposal Lifecycle

1. **Draft**: Any staker submits a Pars Improvement Proposal (PIP) with 1,000 PARS bond.
2. **Discussion**: 14-day community discussion period with structured feedback.
3. **Temperature check**: Non-binding signal vote with 5% quorum.
4. **Formal vote**: Coercion-resistant vote with 20% quorum and 66% supermajority.
5. **Timelock**: 48-hour delay before execution (Guardians may veto).
6. **Execution**: On-chain execution via governance contract.

6 Zero-Knowledge Identity

Identity in Pars Network is built on zero-knowledge proofs enabling selective disclosure without revealing underlying personal data.

6.1 Identity Claims

Definition 12 (Identity Claim)**.** *A tuple* $(c, v, \sigma_{issuer}, \pi)$ *where:*

- c*: Claim type (e.g., "age_over_18", "country_of_origin", "language_proficiency")*
- v*: Claim value (committed, not revealed)*
- σ_{issuer}*: Issuer's signature on the commitment*
- π*: Zero-knowledge proof of the claim's validity*

6.2 Supported Claim Types

Table 4: Identity claim types for diaspora use cases

Claim Type	Description	Proof System
`age_range`	Age within a range without revealing exact date	Range proof
`nationality`	Set membership proof for nationality	Set membership
`language`	Language proficiency attestation	Signature verification
`residency`	Current country of residence	Set membership
`diaspora_member`	Verified diaspora community member	Group signature
`professional`	Professional credential verification	Credential proof
`reputation`	Accumulated network reputation score	Range proof
`stake_tier`	Staking tier without revealing exact amount	Range proof

6.3 ZK Circuit Design

Identity proofs are implemented using Groth16 SNARKs [10] over BN254:

Algorithm 5 ZK Identity Proof Generation

Require: Claim (c, v), issuer signature σ, predicate P
Ensure: Proof π that $P(v) = \text{true}$ without revealing v

```
comm ← Pedersen(v, r)                          ▷ Commitment to value
Construct circuit C:
    Verify σ is valid signature on comm under issuer's key
    Verify P(v) = true
    Verify comm = Pedersen(v, r)
π ← Groth16.Prove(CRS, C, (v, r, σ))
return (π, comm)
```

6.4 Anonymous Credential System

Building on CL signatures [11], we construct an anonymous credential system where:

Property 1 (Unlinkability)**.** *Multiple presentations of the same credential cannot be linked to each other or to the issuance event.*

Property 2 (Selective Disclosure)**.** *The credential holder can prove arbitrary predicates over credential attributes without revealing the attributes themselves.*

Property 3 (Issuer Privacy)**.** *The issuer's identity can be hidden within an anonymity set of authorized issuers using ring signatures.*

7 Post-Quantum Migration

Pars Network implements a hybrid cryptographic strategy providing immediate classical security with a clear path to full post-quantum security.

7.1 Hybrid Scheme Design

Definition 13 (Hybrid Construction)**.** *For any cryptographic operation $\mathcal{O}$, the hybrid scheme computes:*

$$result = \mathcal{O}_{classical}(input) \oplus_{KDF} \mathcal{O}_{pq}(input)$$

where $\oplus_{KDF}$ *denotes combination through a key derivation function. Security is maintained if* either *the classical or post-quantum component remains secure.*

7.2 Algorithm Selection

Table 5: Cryptographic algorithm selection

Function	Classical	Post-Quantum	Standard
Key Exchange	X25519	Kyber-1024	NIST FIPS 203
Signatures	Ed25519	Dilithium-3	NIST FIPS 204
Hash-based Sig	—	SPHINCS+-256f	NIST FIPS 205
Symmetric	AES-256-GCM	AES-256-GCM	NIST FIPS 197
Hash	SHA-3-256	SHA-3-256	NIST FIPS 202
KDF	HKDF-SHA256	HKDF-SHA3-256	RFC 5869

7.3 Migration Timeline

1. **Phase 1** (Current): Hybrid classical + PQ for key exchange and signatures.
2. **Phase 2** (2027): PQ-only mode available as opt-in for high-security users.
3. **Phase 3** (2028): PQ-only mode becomes default; classical fallback retained.
4. **Phase 4** (2029): Classical-only mode deprecated; PQ required.

7.4 Performance Impact

Table 6: Hybrid vs. classical performance overhead

Operation	Classical	Hybrid	Overhead
Key generation	0.04 ms	0.12 ms	3.0×
Key exchange	0.08 ms	0.31 ms	3.9×
Sign	0.05 ms	0.48 ms	9.6×
Verify	0.06 ms	0.19 ms	3.2×
Signature size	64 B	2,484 B	38.8×
Public key size	32 B	1,984 B	62.0×

The signature size increase is the primary concern. We mitigate this through:

- Batch verification: Aggregating multiple Dilithium signatures reduces per-signature overhead by 40%.
- Selective application: Only cross-epoch and governance signatures use hybrid mode; intra-epoch messages use classical signatures with periodic PQ anchoring.
- Compression: Dilithium signatures compress by approximately 15% using the NTT representation.

8 Economics and Incentives

8.1 Token Model

The PARS token serves three functions:

1. **Gas**: Transaction fees on the rollup layer, denominated in PARS.
2. **Staking**: Validators and relays stake PARS for participation rights.
3. **Governance**: Voting weight proportional to staked PARS (with quadratic scaling).

8.2 Supply Schedule

Table 7: PARS token allocation

Category	Allocation	Percentage	Vesting
Community treasury	300,000,000	30%	10-year linear
Staking rewards	250,000,000	25%	20-year decay
Ecosystem grants	150,000,000	15%	5-year linear
Core contributors	100,000,000	10%	4-year, 1-year cliff
Initial liquidity	100,000,000	10%	Unlocked at genesis
Foundation reserve	100,000,000	10%	10-year linear
Total supply	**1,000,000,000**	**100%**	

8.3 Fee Model

Transaction fees follow an EIP-1559-style mechanism adapted for diaspora economics:

$$\text{fee}(t) = \text{base_fee}(t) + \text{priority_fee}(t) \tag{5}$$

where the base fee adjusts according to block utilization:

$$\text{base_fee}(t+1) = \text{base_fee}(t) \cdot \left(1 + \frac{1}{8} \cdot \frac{\text{gas_used}(t) - \text{gas_target}}{\text{gas_target}}\right) \tag{6}$$

Base fees are burned. Priority fees are distributed to the sequencer. A *diaspora discount* of 20% applies to identity-verified accounts, incentivizing identity credential adoption.

8.4 Relay Incentives

Relay nodes earn rewards proportional to their contribution:

$$\text{reward}(r, e) = R_e \cdot \frac{\text{bandwidth}(r, e) \cdot \text{uptime}(r, e)}{\sum_{r' \in \mathcal{R}} \text{bandwidth}(r', e) \cdot \text{uptime}(r', e)} \tag{7}$$

where R_e is the total relay reward budget for epoch e.

8.5 Economic Sustainability Analysis

Table 8: Projected network economics at various scales

Metric	Year 1	Year 3	Year 5
Active users	50,000	500,000	2,000,000
Daily transactions	100,000	2,000,000	10,000,000
Avg. fee (USD)	$0.02	$0.01	$0.005
Daily fee revenue	$2,000	$20,000	$50,000
Annual fee revenue	$730,000	$7,300,000	$18,250,000
Validator APY	8.5%	6.2%	4.8%
Relay APY	12.0%	8.5%	6.0%

9 Cultural Infrastructure

Pars Network is not merely a technical system but a cultural one. The following design decisions reflect this priority.

9.1 Persian-First Design

1. **Calendar**: The Solar Hijri calendar (Jalali) is the default time representation. All on-chain timestamps include both Unix epoch and Jalali date. The `CALENDAR_CONVERT` precompile enables smart contracts to perform date arithmetic in Jalali.

2. **Text handling**: All protocol messages support UTF-8 with proper BiDi (bidirectional) text handling. The `RTL_HASH` precompile ensures deterministic hashing of right-to-left text by normalizing Unicode before hashing.

3. **Name system**: The Pars Name Service (PNS) supports `.pars` domains in both Latin and Persian script. Punycode conversion is handled at the protocol level, not the application level.

4. **Governance language**: All PIPs must include Persian-language summaries. Voting interfaces default to Persian with automatic translation.

9.2 Diaspora Bridge Services

1. **Remittance corridors**: Optimized payment channels for the top five diaspora remittance corridors (US→IR, DE→IR, CA→IR, AE→IR, UK→IR).

2. **Document attestation**: On-chain notarization of translated documents with ZK proofs of authenticity.

3. **Professional credentials**: Verifiable credential bridges between Iranian and host-country professional licensing systems.

4. **Community registry**: Privacy-preserving directory of diaspora community organizations with verified contacts.

10 Evaluation

We evaluate Pars Network across three dimensions: rollup performance, Session layer latency, and cryptographic overhead.

10.1 Testnet Configuration

Table 9: Testnet configuration

Parameter	Value
Validator nodes	21
Relay nodes	100
Light nodes	1,000
Geographic distribution	5 continents, 15 cities
Block time	1.2 seconds
Block gas limit	30M gas
Session message size	4 KB average
Test duration	72 hours

10.2 Rollup Performance

Table 10: Rollup throughput benchmarks

Transaction Type	TPS	Latency (p50)	Latency (p99)
Simple transfer	2,400	1.1s	2.3s
ERC-20 transfer	1,800	1.2s	2.5s
Identity verification	800	1.5s	3.1s
Governance vote	600	1.8s	3.8s
Smart contract deploy	200	2.1s	4.5s

10.3 Session Layer Performance

Table 11: Session layer messaging benchmarks

Operation	Latency (p50)	Latency (p99)	Bandwidth
Session establishment	245 ms	680 ms	12 KB
1:1 message delivery	89 ms	310 ms	4.2 KB
Group message (100)	120 ms	450 ms	4.5 KB
Group message (1,000)	340 ms	890 ms	5.1 KB
Key rotation	180 ms	520 ms	8.4 KB
Offline retrieval	450 ms	1,200 ms	varies

10.4 ZK Proof Performance

Table 12: Zero-knowledge proof benchmarks

Proof Type	Prove	Verify	Proof Size	Constraints
Age range	1.2s	4 ms	192 B	12,000
Nationality set	2.1s	4 ms	192 B	25,000
Credential ownership	3.4s	5 ms	192 B	48,000
Vote validity	4.8s	6 ms	256 B	72,000
Reputation range	1.8s	4 ms	192 B	18,000

10.5 Comparison with Existing Solutions

Table 13: Comparison with existing diaspora and privacy platforms

Feature	Pars	Signal	Zcash	Status
E2E encryption	✓	✓	N/A	✓
Smart contracts	✓	—	Limited	✓
Coercion-resistant governance	✓	—	—	—
ZK identity	✓	—	Partial	—
Post-quantum	Hybrid	—	—	—
Persian-first UX	✓	—	—	—
Diaspora services	✓	—	—	—
Censorship resistance	High	Medium	Medium	Medium
Deniability	✓	✓	N/A	Limited

11 Related Work

11.1 Censorship-Resistant Communication

Signal Protocol [7] provides the foundation for modern encrypted messaging, and our Session Protocol extends it with post-quantum ratcheting and deniable authentication suitable for diaspora threat models. Tor [12] provides anonymity at the network layer; Pars integrates Tor as a transport option while adding application-layer privacy through the Session daemon. Briar [13] demonstrates mesh networking for censored environments; we adopt similar principles for our relay layer.

11.2 Privacy-Preserving Blockchains

Zcash [14] pioneered ZK-SNARKs for transaction privacy. Monero [15] uses ring signatures for sender privacy. Aztec [16] brings private transactions to Ethereum. Pars builds on these foundations but uniquely combines transaction privacy with identity privacy and governance privacy in a single coherent system.

11.3 Decentralized Identity

W3C DIDs [17] provide the standard for decentralized identifiers. Verifiable Credentials [18] enable portable attestations. We extend these standards with diaspora-specific claim types and ZK proof systems for selective disclosure.

11.4 Coercion-Resistant Voting

Juels, Catalano, and Jakobsson [8] introduced the JCJ protocol for coercion resistance. Civitas [9] provided a practical implementation. Our contribution extends these with blockchain-based tallying and panic credentials.

12 Limitations and Future Work

1. **Metadata leakage**: While message content is encrypted, network-level metadata (IP addresses, timing) requires Tor/I2P integration, which adds latency. A dedicated mixnet is planned for Phase 2.

2. **Key management complexity**: The four-level key hierarchy, while necessary for security, creates UX challenges. Research into threshold key management and social recovery is ongoing.

3. **Regulatory uncertainty**: Operating across 60+ jurisdictions creates compliance complexity. The legal framework for privacy-preserving identity across sanctions regimes remains unsettled.

4. **Scalability ceiling**: Current testnet performance (2,400 TPS) may be insufficient for mass adoption. ZK-rollup migration is under investigation for Phase 3.

5. **Post-quantum ZK proofs**: Current Groth16 proofs are not post-quantum secure. Migration to lattice-based proof systems (e.g., STARK-based) is a priority research direction.

13 Conclusion

Pars Network addresses a fundamental gap in digital infrastructure for diaspora communities. By combining an EVM-compatible rollup for programmable settlement with a privacy-preserving Session layer for communication and governance, we provide a unified platform that satisfies the unique security requirements of communities operating under authoritarian threat models.

The key innovations—hybrid post-quantum cryptography, coercion-resistant governance with panic credentials, zero-knowledge diaspora identity, and Persian-first cultural infrastructure—are not merely academic contributions but direct responses to the lived experience of the Iranian diaspora.

Our testnet results demonstrate that these security properties can be achieved without prohibitive performance costs: 2,400 TPS on the rollup layer, sub-second message delivery, and practical ZK proof generation times confirm the system's viability for production deployment.

We invite the broader cryptography and distributed systems communities to collaborate on this work. The challenges faced by the Persian diaspora—surveillance, censorship, financial exclusion, identity fragmentation—are shared by diaspora communities worldwide. Pars Network aims to be not only a solution for one community but a template for sovereign diaspora infrastructure globally.

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A Formal Security Proofs

A.1 Deniability Proof for PSP

Theorem 2 (Full Deniability)**.** *The Pars Session Protocol achieves offline deniability against a judge who receives the full protocol transcript, all participants' long-term keys, and the message plaintext.*

Proof. We construct a simulator $\mathcal{S}$ that, given Alice's identity key ik_A and Bob's identity key ik_B, produces a transcript indistinguishable from a real execution.

Step 1. $\mathcal{S}$ generates ephemeral keys $ek'_A \leftarrow \text{KeyGen}()$.

Step 2. $\mathcal{S}$ computes the four DH values using its knowledge of both identity keys:

$$dh'_1 = ik_A^{spk_B} \quad \text{(computable by Bob)} \tag{8}$$

$$dh'_2 = ek_A'^{ik_B} \quad \text{(computable by Alice)} \tag{9}$$

$$dh'_3 = ek_A'^{spk_B} \quad \text{(computable by either)} \tag{10}$$

$$dh'_4 = ek_A'^{epk_B} \quad \text{(computable by either)} \tag{11}$$

Step 3. For the Kyber component, $\mathcal{S}$ generates a fresh encapsulation (ct', ss'_{pq}) using Bob's Kyber public key. Since Kyber is IND-CCA2 secure, the simulated ciphertext is indistinguishable from a real one.

Step 4. $\mathcal{S}$ derives session keys from $\text{KDF}(dh'_1 \| dh'_2 \| dh'_3 \| dh'_4)$ combined with ss'_{pq} and encrypts the known plaintext.

The resulting transcript $(ek'_A, ct', \text{ciphertext}')$ is computationally indistinguishable from a real transcript under the DDH assumption and Kyber IND-CCA2 security. Any party with access to both identity keys can produce an equally valid transcript, establishing deniability. □ □

A.2 Coercion Resistance of Panic Credentials

Theorem 3 (Panic Credential Indistinguishability)**.** *Given a real credential σ_r and a panic credential σ_p for voter v, no probabilistic polynomial-time adversary $\mathcal{A}$ can distinguish between them with advantage greater than $negl(\lambda)$.*

Proof. Both σ_r and σ_p are CL signatures [11] on commitments to different credential values. Under the LRSW assumption, CL signatures are unforgeable and zero-knowledge. The rerandomization in Algorithm 4 (Step 1) produces fresh, unlinkable presentations. Since the credential

values are committed using Pedersen commitments and revealed only inside the ZK proof, $\mathcal{A}$ would need to break the hiding property of Pedersen commitments (computationally equivalent to DLP) to distinguish the credentials. The reduction is standard. □ □

B Protocol Parameters

Table 14: Complete protocol parameter set

Parameter	Description	Value
λ	Security parameter	128 bits
n_{val}	Minimum validator count	21
n_{relay}	Target relay count	1,000
τ_b	Block time	1.2 s
τ_c	Challenge period	7 days
τ_e	Epoch length	256 blocks
B_{min}	Minimum validator bond	32,000 PARS
B_{relay}	Minimum relay bond	1,000 PARS
Q	PQ ratchet frequency	50 messages
N_{epk}	Ephemeral pre-key batch size	100
τ_{spk}	Signed pre-key rotation	30 days
G_{target}	Block gas target	15M
G_{limit}	Block gas limit	30M
q_{temp}	Temperature check quorum	5%
q_{formal}	Formal vote quorum	20%
θ_{pass}	Passing threshold	66%
τ_{lock}	Governance timelock	48 hours

C Glossary

BiDi
Bidirectional text rendering algorithm for mixed LTR/RTL scripts.

CL Signature
Camenisch-Lysyanskaya signature scheme supporting anonymous credentials.

Double Ratchet
Key management algorithm providing forward secrecy and post-compromise security.

Groth16
Zero-knowledge proof system with constant-size proofs and fast verification.

Jalali Calendar
Solar Hijri calendar used in Iran and Afghanistan.

Kyber
Lattice-based key encapsulation mechanism, NIST PQC standard.

Dilithium
Lattice-based digital signature scheme, NIST PQC standard.

PIP Pars Improvement Proposal, the governance mechanism for protocol changes.

PNS
Pars Name Service, the decentralized naming system for `.pars` domains.

PSP
Pars Session Protocol, the encrypted communication protocol.

SPHINCS+
Hash-based signature scheme, NIST PQC standard.

VRF
Verifiable Random Function, used for unpredictable but verifiable randomness.

X3DH
Extended Triple Diffie-Hellman key agreement protocol.